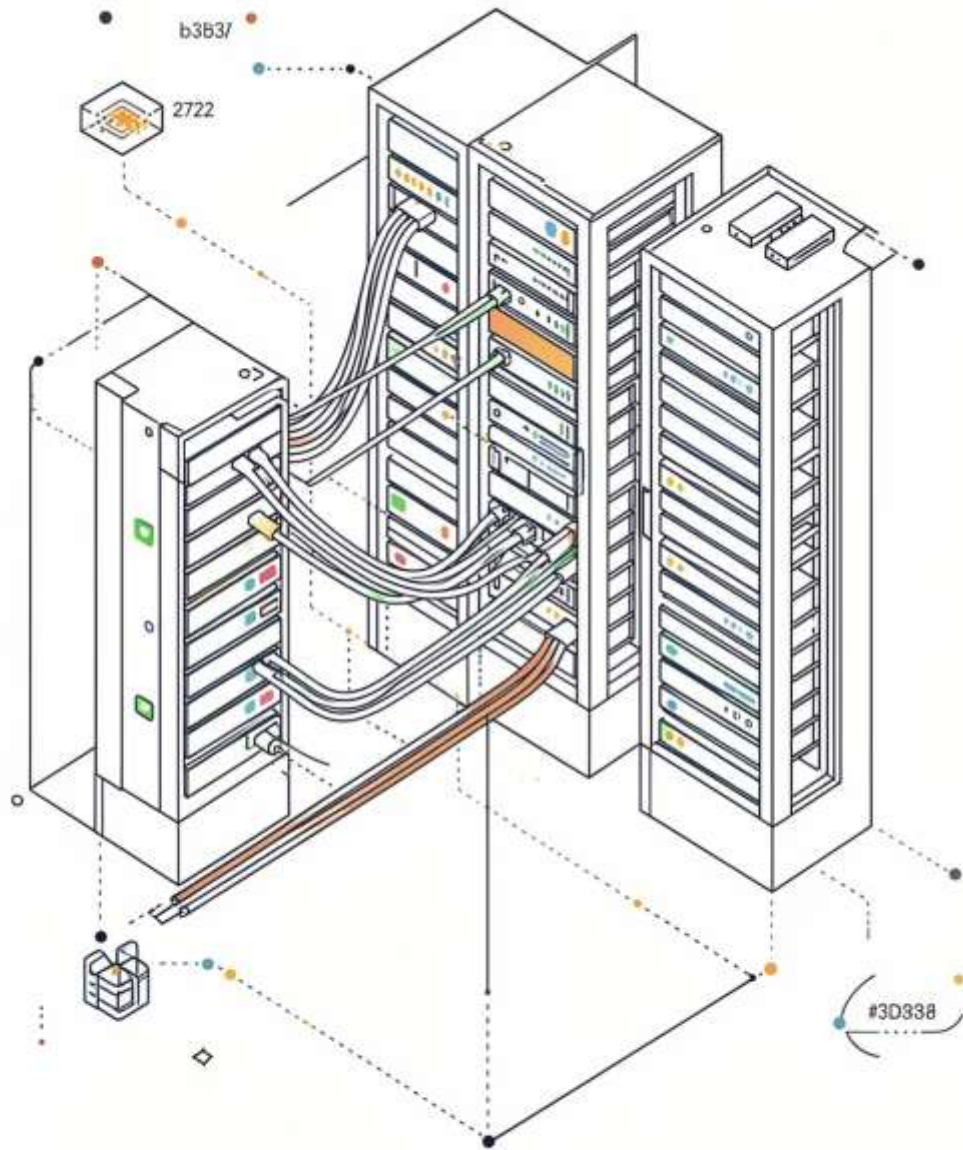




# Intelligent Network Analyser

*A hybrid cybersecurity assessment platform bridging external reconnaissance and internal LAN surveying — built for the modern threat landscape.*

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# The Problem

## Fragmented Tools

*External and internal scanners work separately, which can create dangerous blind spots.*

## SME Gap

*Enterprise solutions are too costly and complex for small and medium enterprises.*

## Manual Overhead

*No automated reporting — compliance documentation is slow and error-prone.*

## Rogue Devices

*Insider threats and shadow IT go undetected without real-time anomaly detection.*



# What is Intelligent Network Analyser ?

## Hybrid Assessment Platform

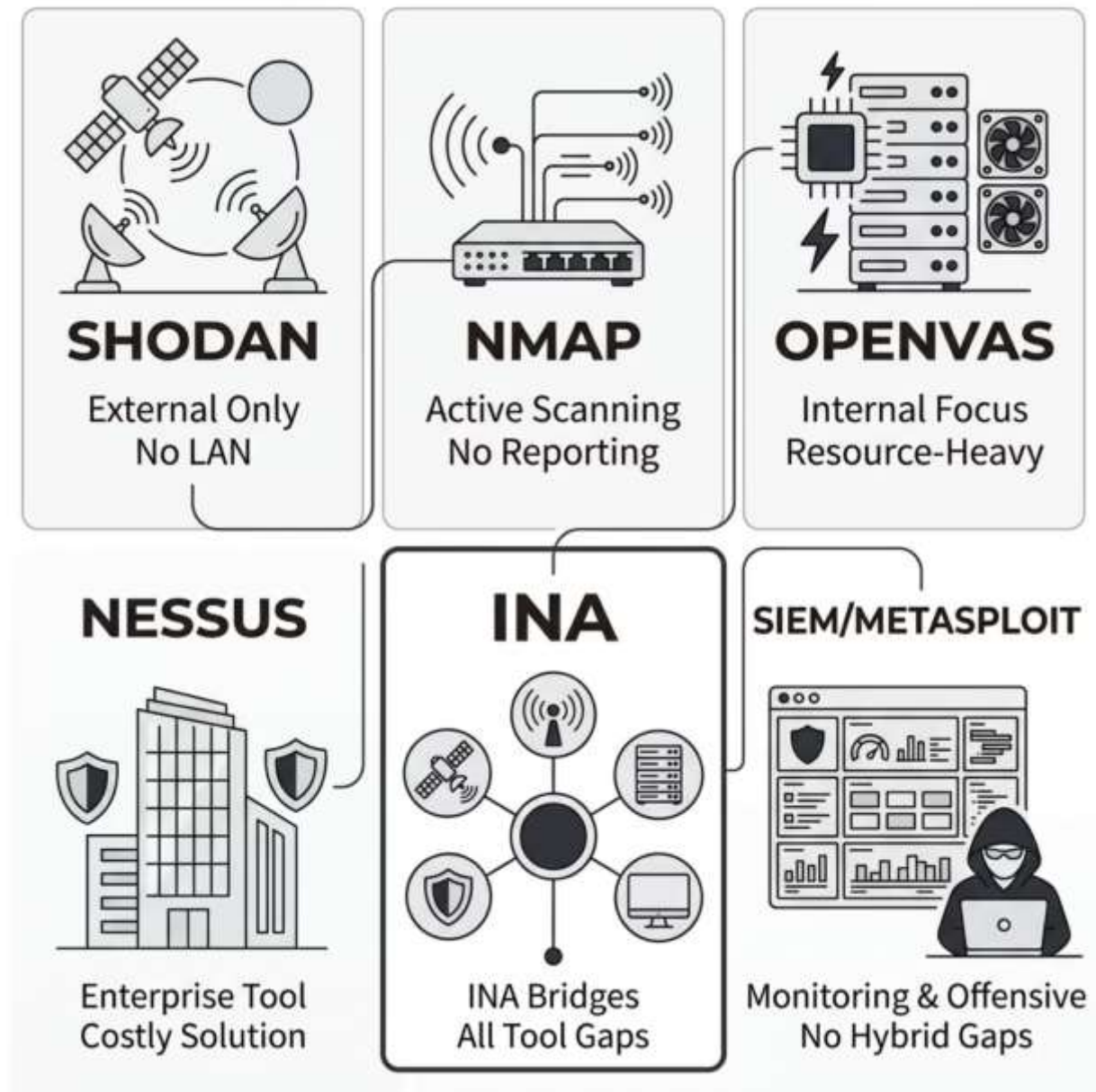
*Built on FastAPI + SQLite, INA unifies:*

- *External Reconnaissance — Shodan + Nmap*
- *Internal LAN Surveying — device inventory + risk scoring*
- *Rogue Detection — anomaly-based, real-time*
- *Automated Reporting — compliance-ready PDF audits*





# Literature Review: The Tool Landscape



## Key Research Gaps

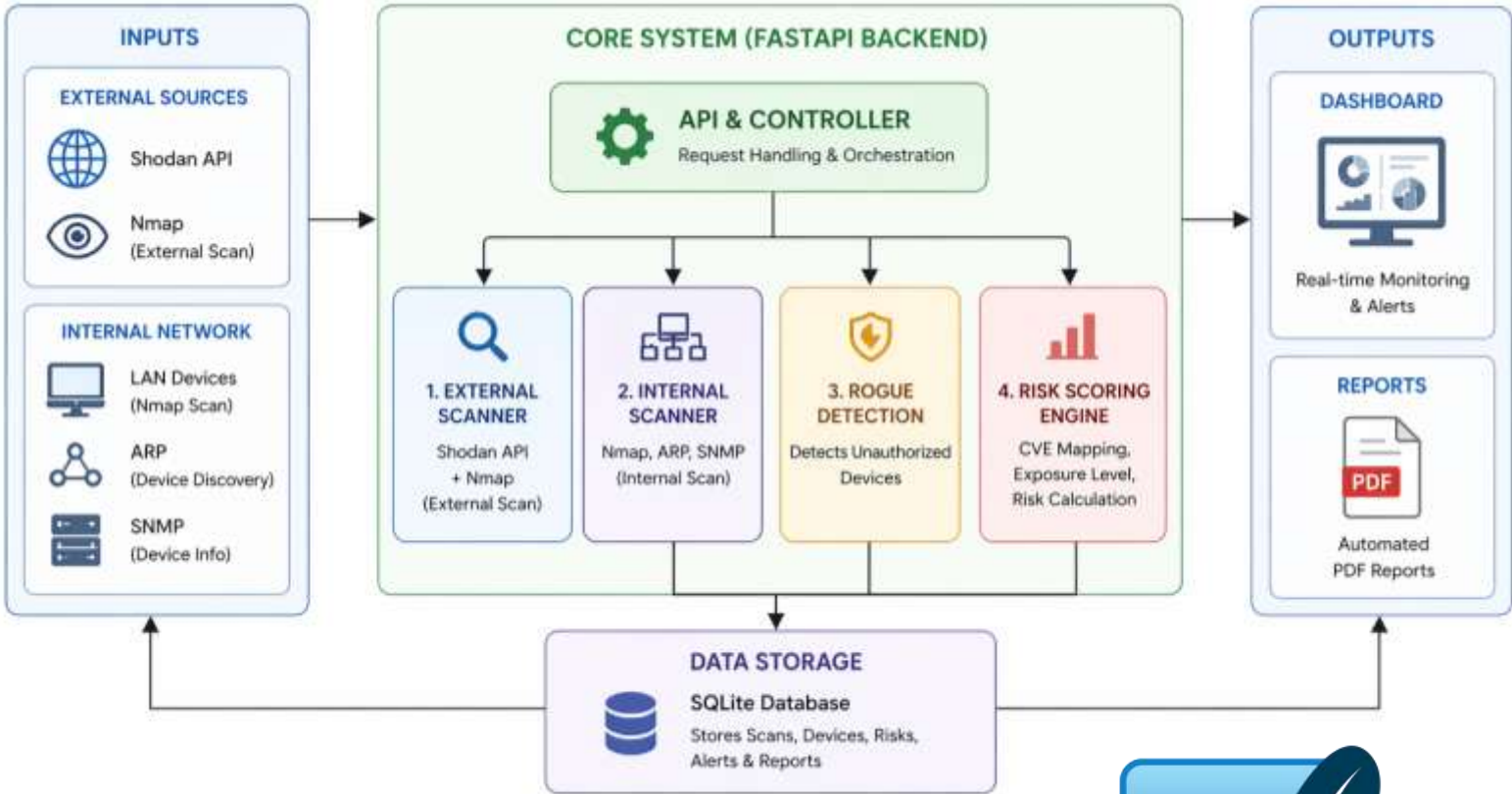
- No tool combines external + internal scanning
- Lightweight options lack depth
- Rogue detection is reactive, not real-time
- Compliance reporting remains manual

□ Intelligent Network Analyser directly responds to all four gaps identified in academic literature.

# System Architecture & Design



## Modular Architecture



### FastAPI Core

*Central controller —  
async, concurrent  
scanning*

### Risk Scoring Engine

*CVSS scores + exposure  
+ anomaly indicators*

### SQLite + SQLAlchemy

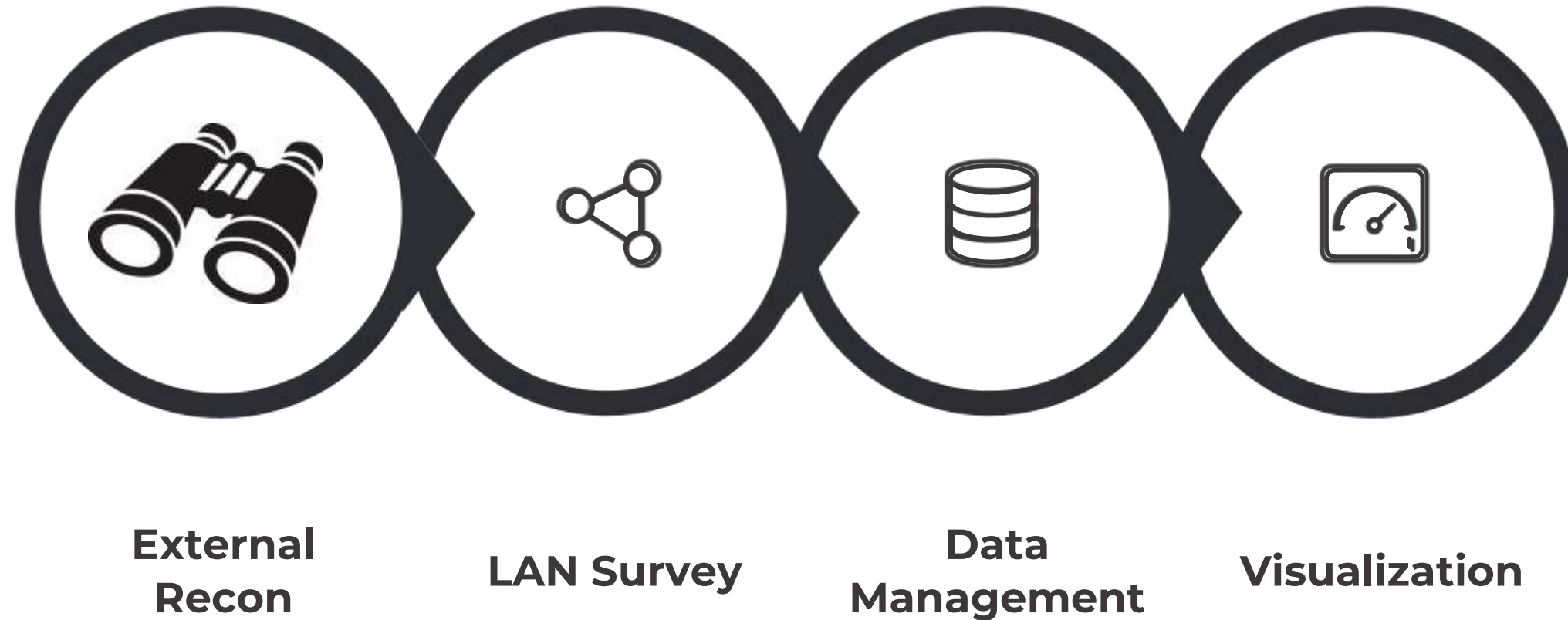
*Lightweight  
persistence, historical  
trend analysis*

### NOC Dashboard + ReportLab

*Real-time visualization  
+ PDF audit reports*



# Methodology: Four-Phase Approach



*Each phase feeds the next — from raw scan data to actionable, compliance-ready intelligence delivered through a unified platform.*

# Implementation & UI

## Key Implementation Steps

01

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### Environment Setup

*Python 3.x, FastAPI, SQLite schema*

02

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### External Module

*Shodan API + python-nmap integration*

03

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### Internal Module

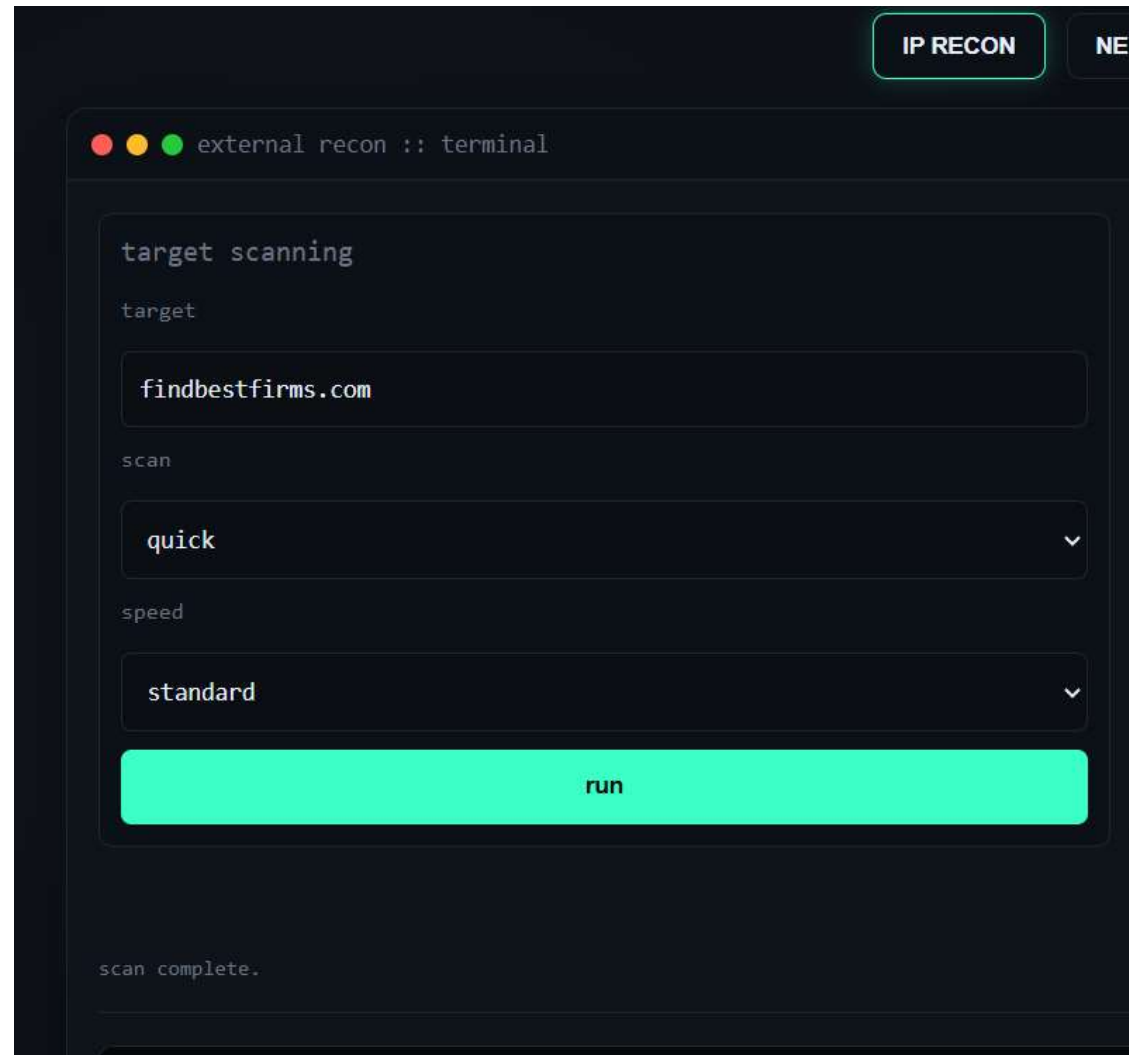
*ARP analysis, CVE-based risk scoring, rogue detection*

04

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### Reporting

*ReportLab PDF generation with exec summaries*



# Testing Results

98%

External Accuracy

Open ports & banners identified via Shodan + Nmap

96%

LAN Detection Rate

48/50 devices detected; IoT fingerprinting improved

30s

Rogue Detection

Unauthorized devices flagged in under 30 seconds

<2min

Report Generation

Compliance-ready PDF with exec summary + findings

200

Max Devices

Stable scans in ~15 min; dashboard remained responsive

✔ All five primary objectives met or exceeded. Rogue detection surpassed expectations — false positives reduced from 12% to 3% after threshold refinement.





# Conclusion & Future Scope

## Intelligent Network Analyzer

- *Hybrid external + internal visibility*
- *Real-time rogue device detection*
- *Automated compliance-ready reporting*
- *Lightweight access for SMEs*

## Future Directions

- *Scalability — distributed scanning for 1000+ devices*
- *AI/ML — predictive threat classification*
- *IoT/OT — ML-based fingerprinting*
- *SIEM/SOAR — enterprise integration*
- *Cloud — extend to VMs and containers*